

Boming Zhang

Backend Software Engineer

San Jose, CA | (706) 363-5179 | [Email](#) | [Website](#) | [GitHub](#) | [LinkedIn](#)

EXPERIENCE

Backend Software Engineer

Jul. 2025 - Present

TikTok Inc.

San Jose, CA

- Led design and delivery of a **privacy compliance platform** (self-designed data model, workflow engine, configurable notifications) achieving **100% delivery success** and fully auditable trail.
- Built **recommendation system infrastructure**: AB experiment platform for holdout studies, model migration pipeline (ERD through AB launch), and transparency controls API for regulatory compliance.
- Led database schema redesign for a **research access platform**; delivered **time-based access-control**, a **rule-based query builder** (~90% dev time reduction), and data upload flow (**30–40 hrs to 1–2 hrs** per request). Migrated services across cloud infrastructures (**5–20 days to 1–2 days** deployment).
- Resolved **5 critical on-call incidents** on a high-traffic platform; built traffic monitoring and alert tooling (**hours to minutes**); executed a **zero-downtime service migration**.
- Owned backend for **creator monetization** and **public safety alerts**: delivered improved donation flow via AB testing, root-caused alert delivery failures (**11 to 0**), and resolved **8+ latent production bugs** post-launch.

Software Engineer (Intern, Full-Time)

Feb. 2022 - Jul. 2022, Nov. 2022 - Jul. 2023

Naive Systems, advisor Xiao Jia, Kevin Sun

Shanghai, China

- Developed an **LLVM-based static analysis product** with **100+ checkers** (**syntax trees, lexical analysis, symbolic execution**); contributed an **LLVM bug report** (**10% false-negative reduction**).
- Optimized multi-process unit testing during CI, achieving a **25x speedup** in test execution time.

Full Stack Development Engineer Intern

Jun. 2021 - Feb. 2022

Shanghai Zhimian Weiye Technology, advisor Prof. Mian Li, Dr. Ruixiang Zheng

Shanghai, China

- Initiated the development of a **high performance firefighting platform (Web + App)** based on IoT data. Used FAHP algorithm to detect potential hazards and report **95% less false fire alarms** in **milliseconds response time**.
- Scaled the system to **over 7M+ IoT devices** with 99.99% uptime; deployed on **the fire brigade of Shanghai Province**, protecting 20M+ residents.

PROJECTS

Joint Online Judge | *Python3, Go, PostgreSQL, Redis, Grafana, Docker, React.js*

Mar. 2021 - Present

- Led a team of **10+ students** to develop and maintain a **distributed online judge system** with analytic tools. Provide online judge service for **1k+ students** and **10k+ submissions monthly**.
- Built secure sandbox using **Linux namespaces and containerization** for process isolation.

AList (Github 50k stars) | *Go, Go Gin, GORM, Solid.js, WebDAV, SMB, S3, FTP*

Nov. 2022 - May. 2023

- 3rd main contributor**, focusing on implementing and optimizing **storage drivers for various protocols**.
- Designed and implemented a high-performance search feature, handling large-scale virtual file systems efficiently.

FOCS student system | *Python3, Go, Node.js, Docker, Nginx, Github Actions*

Nov. 2018 - Present

- Managed an academic platform with online judge, Git service, and CI for **2k+ students across 10 courses** to build, test, and grade code, while architecting and maintaining critical infrastructure.
- Developed CLI tools for system automation, garnering 200+ monthly downloads on PyPI.

TECHNICAL SKILLS

Languages: Python3, Go (golang), C, (modern) C++, LLVM IR, JavaScript, Assembly, Shell, SQL

Systems & Tools: Linux, Git, Docker, LLVM, GDB, Valgrind, Wireshark, GNU Make, CMake, opencode, Cursor

Frameworks & Libraries: FastAPI, Django, SQLAlchemy, Go Gin, GORM, React.js, Svelte, Protocol Buffers

Databases & Services: PostgreSQL, MongoDB, Redis, Nginx, GitHub Actions, Grafana, Jenkins

EDUCATION

The University of Georgia (School of Computing)

Aug. 2023 - May. 2025

MSc in Computer Science (GPA 3.92/4.0)

Athens, GA

- Relevant Coursework: Software Engineering (A), Privacy-Preserving Data Analysis (A), Data Mining (A)

Shanghai Jiao Tong University (UM-SJTU Joint Institute, ABET acred.)

Sept. 2018 - Aug. 2022

BEng in Electrical and Computer Engineering (minor in Computer Science, GPA 4.0/4.0)

Shanghai, China

- 2020-2021 Undergraduate Excellent Scholarship.** Relevant Coursework: Operating Systems (A+), Computer Architecture (A), Data Structures and Algorithms (A)